

Machine-Readable Economics Papers for Agentic Research

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Abstract

AI research systems read papers to retrieve literature, evaluate manuscripts, write code, and attempt reproduction, but the papers themselves remain mostly PDFs: readable by people, but hard to query as evidence. This paper builds an economics-specific pipeline that converts full economics-paper PDFs, including in-PDF appendices, into structured records linked to their source locations. Machine-readable here means more than LLM-ready text: each paper becomes records that can be stored in databases, queried directly, compared across papers, analysed with standard algorithms, and selectively returned to an LLM with source evidence. The pipeline extracts seven information types: data sources, methods, findings, limitations, open questions, citation evidence, and replication metadata. The same records are exposed as SQL tables, semantic-search indexes, and graph objects. The result is both a queryable database of paper evidence and a dataset about the structure of economics research. Applied to 1,608 selected 2025–2026 NBER working papers, the first build shows that this representation can be populated beyond a small coded sample and used for source-grounded search, cross-paper measurement, and graph-based research workflows.¹

Keywords: machine-readable papers; economics literature; source-linked extraction; knowledge graphs; hybrid search; retrieval-augmented generation; AI research agents; research infrastructure.

JEL codes: C55, C81, C88, D83, O33.

1 Introduction

AI research tools increasingly retrieve literature, evaluate manuscripts, write code, and attempt empirical reproduction from papers (Kohler et al., 2026; Zheng et al., 2025). The scientific object these systems need is the paper, but most papers remain PDFs: readable by people, hard to query as evidence. The bottleneck is not unique to economics, but economics is a useful first setting because working papers are central research objects and their evidence often depends on datasets, methods, estimates, identification designs, and appendix material that are not visible from metadata or abstracts alone. This paper treats the problem as one of representation, not

*Claude Code (Opus 4.6 and 4.7) and Codex (GPT-5.4 and GPT-5.5) were used to assist with drafting, editing, coding, extraction review, table and figure preparation, and cluster-label review. I reviewed the outputs and take responsibility for all statements, analysis, and errors.

¹The release reports coverage descriptives and a preliminary single-reviewer validation review in Appendix G; future versions will improve automatic promotion, validate source spans and typed links, and benchmark retrieval and downstream tasks before larger-corpus release.

only retrieval: economics papers need to be converted into structured research objects that can be queried, compared, and audited. Existing tools often treat papers as retrievable documents; this paper treats structured records over papers as the unit that researchers can search, measure, link, and audit.

The paper builds this layer for the 2025–2026 NBER working-paper corpus. The pipeline reads the full working-paper PDF, including appendices contained inside the PDF, and extracts seven kinds of paper information: data sources, methods, findings, limitations, open questions, citation evidence, and replication metadata. Each item keeps its source location: page, section, table, figure, equation, or passage. The same records are exposed through structured JSON, SQL tables, semantic-search indexes, and graph objects, so a researcher or agent can combine structured filters, keyword search, semantic retrieval, and link-following before returning to the source evidence. When a working paper later has a published version, this build still processes the NBER working-paper PDF; published versions enter only when they are cited by a processed paper or when a cited-work resolver links a reference to a published DOI. Existing systems already cover important parts of this space. Full-text search is an important first step: it lets researchers search across papers and return to the relevant passages, tables, figures, or sections. The layer built here is complementary: it converts economics papers into source-linked records for methods, data sources, findings, limitations, open questions, citations, and replication metadata. Those records can support search, but they can also serve as a data source for measuring patterns across economics papers. The two uses reinforce each other: structured extraction makes search more precise by turning recurring paper concepts into typed, source-linked records, while the database makes those records reusable across many later queries rather than one-off prompt outputs. Retrieval-augmented agents can search and cite full-text passages, and scholarly knowledge-graph projects such as ORKG show how research contributions can be made machine-actionable (Jaradeh et al., 2019; Lála et al., 2023). Section 2 and Appendix A locate this paper relative to those systems.

The economics-specific part is the ontology and provenance layer, not the PDF format itself. In practical terms, the pipeline gives each paper a field-specific coding guide: different ways of describing the same method, dataset, citation role, limitation, or open question are mapped to common labels, while each label remains linked to the source passage and extraction version.² The output should therefore be read both as infrastructure and as data: a database for querying paper evidence, and a structured dataset for studying the contents of the economics literature. The extraction layer also changes what search can mean: keyword search depends on exact wording, embeddings recover semantic similarity, and LLM preprocessing maps varied paper language onto controlled labels such as method family, dataset type, citation role, or limitation type. The broader contribution is to turn a repeated manual task—reading papers to code methods, datasets, findings, limitations, citations, and replication information—into reusable research infrastructure. For agents, the same layer augments retrieval with domain-specific evidence and provides guardrails by comparing proposed claims, methods, or checks against documented practice in the literature.

The paper makes three contributions. First, it proposes and implements an economics-specific structure for machine-readable papers: seven linked information types, source locations, controlled-vocabulary labels, and typed links among extracted objects. Second, it turns full-text papers

²Technically, this coding guide is a controlled vocabulary and ontology used with LLM extraction. It behaves less like a fixed dictionary of trigger phrases and more like a controlled vocabulary with thesaurus-like behavior: categories and canonical labels set boundaries, while the model can recognise paraphrases, abbreviations, and field-specific variants in context. The source and contract metadata record the prompt bundle, taxonomy, schema, model binding, run configuration, and source location, so a later error can be traced to the relevant extraction component and repaired locally before revalidation. On controlled vocabularies as semantic infrastructure for reconciling synonyms and variants, see Talisman (2025); on workflow ontology design and governance, see Talisman (2026).

into a cross-paper dataset and measurement layer. This is complementary to nearby economics text work. Goldsmith-Pinkham (2026) measures broad method trends with full-text keyword search; Garg and Fetzer (2026) extract a targeted claim graph of standardized concept nodes, claim edges, and evidentiary basis; and Briggs et al. (2026) code null-result reporting. This paper extracts core paper elements and preserves the relationships among them, so the same preprocessing can support generalisable analysis across several units: papers, paper objects, method–data pairs, finding–method links, limitation–open-question links, citation contexts, and corpus-level variables. Once findings, methods, datasets, disagreements, stated limitations, open questions, information gaps, and citation contexts are recorded as typed objects, they can be searched as text, aggregated as variables, or assembled into knowledge-graph inputs. Third, the paper applies the method to the 2025–2026 NBER corpus, reporting representation-coverage descriptives, a preliminary validation review, and four task setups—living systematic review, automated diagnostics, agentic reproduction, and AI peer review—designed for future benchmarking rather than benchmarked here.

The 2025–2026 NBER extraction is the first corpus run. Its descriptives focus on whether the representation is populated across papers, not on substantive economic findings. A first single-reviewer validation review in Appendix G diagnoses the current extraction quality for the main non-ML paper-object layers in a 100-paper sample after adding matched intermediate records; raising recall, automating promotion into final tables, and validating spans, links, and retrieval are the engineering priorities before larger-corpus release. The paper proceeds as follows. Section 2 motivates the layer. Section 3 defines the extraction method and paper objects. Section 4 describes the access modes: structured filters, keyword search, semantic search, and graph or edge traversal. Section 5 reports representation coverage in the 2025–2026 NBER run, and Appendix G reports the preliminary validation review. Section 6 sets out four illustrative downstream task setups for future benchmarking. Section 7 states the current scope, cost, and scaling path. Section 8 concludes.

2 Why AI Research Tools Need Machine-Readable Papers

Recent AI research tools show a common input problem. Across augmented and automated systems, papers are used to retrieve related work, extract claims and equations, check methods, attempt reproduction, generate code, and produce or evaluate manuscripts (Zheng et al., 2025). This paper adds an upstream layer for economics: source-grounded paper records that agents and researchers can query, compare, or check against. Figure 1 organises four task families where that layer is useful. The same representation supports researcher-facing augmentation—search, checking, comparison, and synthesis—and more autonomous workflows that carry tasks across the research cycle, from information gathering through hypothesis development, experimentation, analysis, and iteration (Lu et al., 2026).

Two adjacent approaches motivate but do not replace this layer. Retrieval-augmented scientific QA systems such as PaperQA retrieve full-text articles, select relevant passages, and generate cited answers (Lála et al., 2023). Scholarly knowledge-graph systems such as ORKG show how research contributions can be made machine-actionable through structured graph records (Jaradeh et al., 2019). The gap here is economics-specific preprocessing: full working-paper PDFs are converted once into reusable records whose methods, data, findings, citations, limitations, open questions, and replication metadata remain linked to their source evidence.

This distinction matters because several current tools can already read a PDF. A PDF reader can produce a task-specific analysis for one manuscript or one query. The object built here is different: a reusable evidence layer over many papers. It helps because (i) the same extracted record can answer many later questions without re-reading the PDF; (ii) answers can be com-

pared across papers because datasets, methods, findings, limitations, citation roles, and source locations use shared fields; (iii) a researcher can audit the exact passage, table, figure, equation, or citation context behind a claim; and (iv) the same records can be used through SQL tables, semantic search, and graph objects rather than only through an LLM transcript.

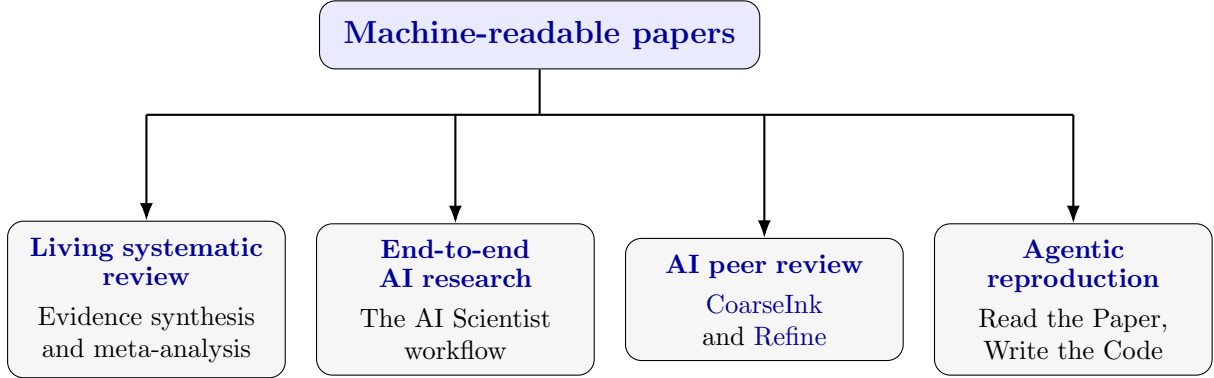


Figure 1: Machine-readable economics papers as reusable input for four AI research tasks (Kohler et al., 2026; Lu et al., 2026; Stocker et al., 2025; Zheng et al., 2025). For each task family, this paper contributes reusable structured paper evidence rather than a new end-user agent or one-off PDF reader: extracted data sources, methods, findings, limitations, open questions, citations, replication information, and source locations. Box titles link to representative source or tool pages.³

The four task families differ, but each needs papers in a structured form rather than only as PDFs or abstracts:

- (i) A living systematic review needs papers that can be screened, coded, updated, and re-analysed as the evidence base changes (Elliott et al., 2014; Iversen & Quintana, 2026).
- (ii) An end-to-end AI research system needs full-paper evidence when it checks novelty, chooses datasets, or chooses methods. In the public AI Scientist workflow, related work is retrieved through scholarly API results that expose metadata and abstracts, not a structured full-paper literature layer (Lu et al., 2026; Sakana AI, 2026).
- (iii) AI peer review needs both submitted-paper evidence and related-literature evidence: what the manuscript itself supports, what comparable papers usually report, and whether important prior work is being used as background, method precedent, comparison, or contradiction.
- (iv) Agentic reproduction needs the paper as a structured description of the study: data, sample, method, tables, equations, and the checks needed to decide whether a reimplementa-tion failure reflects the agent, the extraction, missing data, or underspecification in the paper. It also needs related-paper context, especially when time has passed since publication, to identify later data treatments, validity checks, and robustness standards that go beyond simply rerunning the original code.

The peer-review case makes the distinction clearest. This paper does not claim that tools such as CoarseInk or Refine cannot analyse a submitted PDF. CoarseInk is publicly inspectable and reads the submitted file before running several review passes; Refine is a deployed review tool, but it is not currently open source, so its internal process is not audited here (CoarseInk, 2026; Refine Technologies, 2026). The contribution here is the evidence layer around the PDF: a

³The first box uses the established term *living systematic review*. It denotes an evidence synthesis that can be updated as new studies enter, including meta-analysis where appropriate, over structured full-paper records rather than only abstracts and bibliographic metadata.

checkable record of claims, tables, equations, references, assumptions, citation roles, related papers, and source locations that can ground, revise, or drop a review comment.

Agentic reproduction needs the original paper as an executable description of the study. *Read the Paper, Write the Code* shows that agents can recover published social-science results from paper-derived method and table descriptions plus the original data, without access to the authors’ code (Kohler et al., 2026). That benchmark addresses same-data reimplementations on papers that I4Replication had classified as fully reproducible; it is distinct from testing whether authors’ conclusions are robust to alternative specifications. A structured literature layer can help with the next step by selecting targeted alternative specifications, data treatments, and validity checks from nearby and later papers, while still leaving substantive judgment to the researcher (Dreber & Johannesson, 2025).

The implication is that parts of a paper can become reusable research components. Here, a component does not mean executable software; it means a source-linked record of a dataset, sample rule, constructed variable, method, table, equation, assumption, limitation, or robustness check that can be retrieved independently while remaining linked to the paper. Code is modular by design. Paper-derived records make the empirical components of a study more modular in a weaker but useful sense: they can be searched, compared, adapted, and checked across tasks without pretending that any component alone settles the substantive question. This is also a middle ground between abstract-only literature search and generating an empirical study from scratch. Systems such as The AI Scientist and APE-style economics pipelines need to choose ideas, datasets, methods, and checks; a structured paper layer can supply reusable components from existing papers that are candidates for adaptation to new research questions or datasets, while preserving the source evidence and uncertainty around each choice (Lu et al., 2026; Sakana AI, 2026; Social Catalyst Lab, 2026). Reproduction projects then become a natural testbed for these components, because they reveal whether the extracted records are specific enough to support reimplementations and whether failures come from the agent, the extraction, missing data, or underspecification in the original paper.

The same structure points to the next stage: literature-grounded robustness reproducibility. Recreate reproduction asks whether an agent can recover a published result from the paper and the same data. Robustness reproducibility asks whether the conclusion survives alternative plausible analytical decisions on the same data (Dreber & Johannesson, 2025). The structured corpus helps define those alternatives from the literature itself: papers already nearby in the literature and, as the corpus is updated, papers published later that use the same dataset, outcome, method family, or identification strategy; validity checks commonly reported for that design; author-stated limitations; and citation contexts where later papers compare, qualify, or contradict the original estimate. This matters most for older papers, where reimplementations is only the first question and later disciplinary practice may imply additional checks before treating the result as current evidence. This follows the logic of specification-curve analysis: the alternatives should be theoretically justified, statistically valid, and non-redundant, not arbitrary perturbations (Simonsohn et al., 2020).

Appendix B expands these task families into robustness, source-tracing, and reuse tables. The main text keeps the focus on the input problem: current tools would benefit from structured paper records with source links before they are used for high-stakes research workflows.

The limit of metadata- and abstract-centric retrieval. Current end-to-end research agents often access prior literature through scholarly metadata APIs such as Semantic Scholar or OpenAlex. In the public AI Scientist implementation, novelty checking and citation insertion query these APIs and show the model titles, authors, venues, years, citation counts, and abstracts where available (Lu et al., 2026; Sakana AI, 2026). Abstracts describe findings but omit the methods, validity tests, data structures, and assumptions that determine whether a result transfers to a new setting. An abstract that mentions “difference-in-differences” does not reveal

whether the parallel-trends test was run, reported, or passed — information the agent needs to judge reusability. Abstract retrieval also biases discovery toward direct thematic matches, suppressing cross-field connections that only become visible when methods and data structures are compared across literatures. Structured extraction addresses both gaps: extracted fields are queryable even when they describe what a paper did *not* do, and matching on methods or data can connect papers that share a design even when they use different vocabulary.

From literature review to living systematic review. The synthesis standard for evidence-based research is not a narrative literature review but a systematic review: pre-specified eligibility criteria, structured search and screening, study appraisal, and data extraction (Elliott et al., 2014; Iversen & Quintana, 2026). A meta-analysis is the quantitative component when comparable effects can be re-analysed; it is not synonymous with the review itself. Its value is to reconcile apparently contradictory findings and quantify “what is known and what is not yet known” (Gurevitch et al., 2018). Static reviews rapidly age: search dates lag publication, author-maintained review files go stale, and new studies can change pooled estimates or the interpretation of a field. *Living* systematic reviews address this by keeping eligibility, extraction, appraisal, and, where relevant, meta-analytic estimates current (Elliott et al., 2014; Iversen & Quintana, 2026).

Systematic Online Living Evidence Summaries extend this logic by framing living evidence products as continuously updated, curated, interrogatable evidence datasets enabled by automation (Hair et al., 2023). This paper borrows that database logic but changes the structure being maintained: rather than only keeping study-level evidence for a review, it decomposes each economics paper into source-linked data, methods, findings, limitations, open questions, citation evidence, and replication information.

The hard part is maintenance. Living reviews require transparent update rules, guardrails against reviewer bias, mechanisms for incorporating revised papers, and reusable outputs. Elliott et al. (2014) already pointed to semi-automation through text mining, data repositories, linked data, and semantic technologies. Later automation reviews show that much of the work has focused on search, screening, and primary-study selection, while extraction and knowledge representation remain harder; where extraction is automated, ontologies are often the mechanism for representing concepts and relationships (de la Torre-López et al., 2023; Van Dinter et al., 2021). Structured full-text extraction supplies the missing input layer. A living review can add new papers and revised working papers, then triage whether they add a new estimate, dataset, identification strategy, validity test, limitation, contradiction, or open question. Routine additions can be integrated into the review database; papers that materially change estimates, introduce new data/method combinations, or qualify a central claim can be flagged for closer human review. The database and graph outputs are useful research products in their own right: SQL tables let reviewers filter, aggregate, and re-analyse extracted information, while RDF/OWL-style graph representations expose shared concepts and relationships through common semantic conventions.⁴ That makes the processed literature reusable by other researchers, not just by the original extraction pipeline.

Citation roles as a quality signal for primary studies. Felizardo et al. (2012) use citation networks to help validate the selection of primary studies in systematic reviews. Citation role adds the semantic layer that a network edge lacks. Rather than recording only whether paper A cites paper B, the extraction records the citation function: background context, foundational

⁴This follows the living-review argument in Elliott et al. (2014): structured research outputs, linked data, and semantic technologies can make systematic-review data reusable beyond the original review and support a broader evidence ecosystem. RDF represents knowledge as subject–predicate–object triples, OWL 2 provides ontology vocabulary and semantics for classes, properties, individuals, and their relationships, and SHACL validates RDF data graphs against shape constraints (Berners-Lee et al., 2001; Cyganiak et al., 2014; Knublauch & Kontokostas, 2017; W3C OWL Working Group, 2012). Here, RDF/OWL-style refers to exporting extracted paper items and relationships in interoperable semantic-web forms, not to using RDF or OWL as the extraction model.

theory, methodological precedent, an effect-size benchmark, or a target of critique (Anderson & Lemken, 2023; Jurgens et al., 2018). This should not replace formal quality appraisal by review authors. It is a complementary signal from future citing authors: when later papers use a study as identification precedent, calibration input, or a result to contradict, they provide a different signal than when they cite it only as background. That distinction can guide snowballing and prioritisation in living reviews, especially when new papers arrive faster than human reviewers can read every PDF.

3 Extraction Method and Paper Objects

This section defines the paper representation used to make the PDF machine-readable in the database-and-query sense described above. The unit is a full working-paper PDF, including appendices contained inside that PDF. Each paper goes through five main stages, with enrichment as a 4.7+ substage before contract gating: reading the PDF and identifying sections and tables; classifying the paper type (empirical, structural, or theoretical); extracting the seven kinds of paper information; linking related items, such as findings to methods and limitations to open questions; and contract gating.⁵ The base path averages under two minutes per paper; enrichment adds roughly 30–80 seconds when triggered. For tables, figures, and equations, the system reads the downloaded working-paper PDF, including appendix tables, figures, and equations that are inside that PDF. Separate online appendices are not yet downloaded and processed as independent documents.

The output is deliberately redundant across access modes. The same extracted paper information is written as source-linked JSON, SQL tables, semantic-search views, and graph objects, so non-LLM tools can filter, aggregate, cluster, link, and traverse the records before any LLM is asked to interpret a retrieved passage. In database form, the records support retrieval, filtering, joins, and source inspection. In dataset form, the same records become observations and variables for analysing papers across the corpus. The indexing step is mechanical rather than interpretive: it takes the contract-gated paper information, source passages, table/figure/equation records, and document sections and creates paper profiles, extracted-item views, individual-field views, nearby-sentence views, broader context windows, relationship views, and claim-evidence bundles. These views are stored in DuckDB. When semantic search is needed, the text is embedded with `text-embedding-3-large`. Appendix H shows the database format and the search views in detail. Figure 2 summarizes the resulting paper representation.

⁵SHACL is used as a contract-checking layer, not as an extraction model. The shapes encode required fields, allowed relationship types, cardinality expectations, and controlled-vocabulary constraints. This catches malformed records before downstream use; it does not certify substantive extraction accuracy.

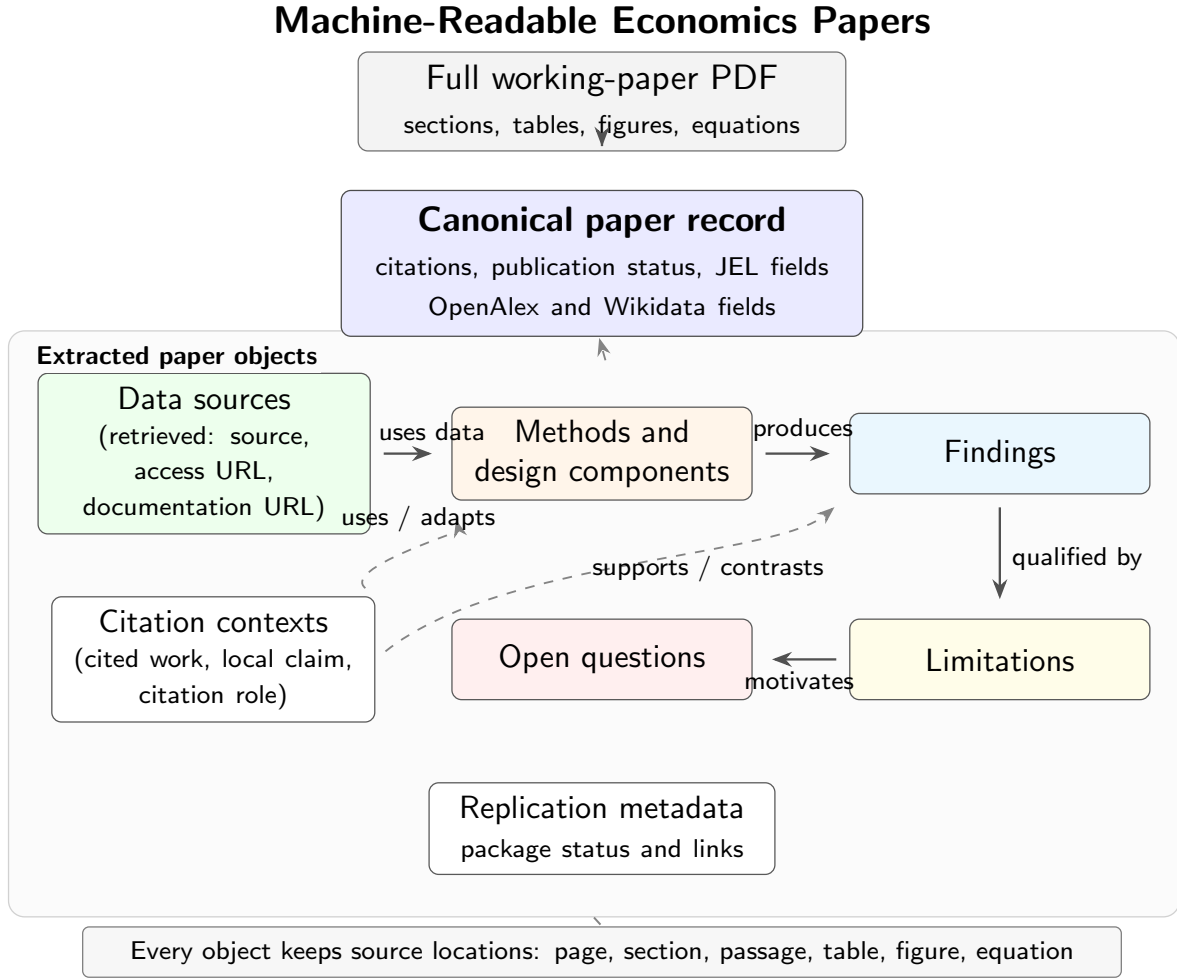


Figure 2: Machine-readable economics paper representation. A full working-paper PDF is converted into a canonical paper record and extracted paper objects. Solid arrows show the main inferential chain from data sources to methods, findings, limitations, and open questions. Dashed arrows show citation-context links and source-location provenance; the full relationship vocabulary is reported in Appendix E.

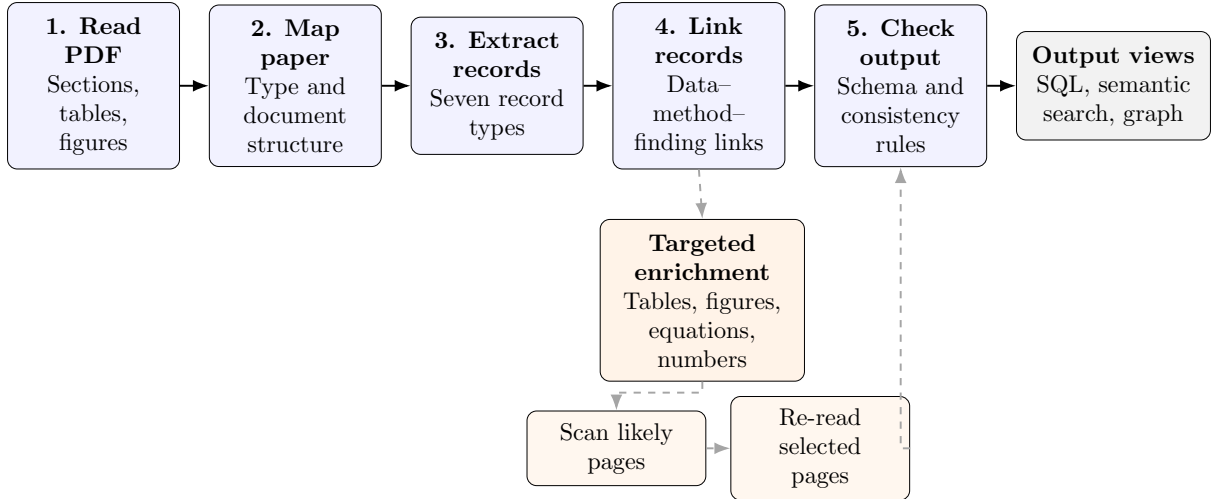


Figure 3: Extraction pipeline for turning one working-paper PDF into searchable paper records. The numbered boxes show the five interpretive stages: read the PDF and detect sections, tables, and figures; map the paper type and document structure; extract the seven record types; link related records such as data, methods, findings, citations, limitations, and replication fields; and check schema and consistency rules. The dashed branch is a targeted enrichment pass for tables, figures, equations, and numeric results: likely pages are scanned and selected pages are re-read before final checks. The gray box writes the checked record to SQL, semantic-search, and graph views without changing the extracted content.

Three properties of the multi-stage design matter for extraction quality. First, each stage has a narrower job: a method stage focuses on methods, while a finding stage focuses on findings. Second, important fields are checked more than once. Methods extracted in Stage 3 are reviewed when links are created in Stage 4, and precision fields such as standard errors and p -values are checked again during enrichment. Third, every extracted item is linked to the page, figure, table, equation, or passage where it appears. If an error reaches downstream use, the user can inspect the source, identify whether the failure came from source parsing, prompt logic, taxonomy mapping, linking, or enrichment, and then update or flag the affected component without re-running the full extraction. This isolation does not remove the need for validation after a fix, but it limits the surface area over which unrelated parts of the pipeline can regress. Full stage-by-stage prompts are in Appendix S.

The seven extracted information types, **data sources**, **methods**, **findings**, **limitations**, **open questions**, **citation evidence**, and **replication metadata**, capture the core inferential chain of an economics paper, from the data that identifies variation, through the methods that exploit it, to the findings that result, the limitations that qualify them, and the open questions they raise (Morgan, 2012). Labels use controlled vocabularies rather than free text wherever possible. Methods are coded at three levels: a broad purpose, displayed in the current run as measurement, causal, descriptive, structural parameter, structural, decomposition, or theoretical; one of 91 method-family labels, and, where needed, embedded method components. Algorithmic work can therefore appear either as the method itself—for example language models, causal forests, double machine learning, or prediction models—or as a component inside another design, such as record linkage inside an event study or a numerical solver inside a structural counterfactual. Shocks are recorded as variation sources inside causal identification designs; counterfactuals are recorded as structural or simulation method families. The remaining controlled vocabularies cover data-collection method categories and citation-function labels (Jurgens et al., 2018). Full controlled-vocabulary tables appear in Appendix R.

- **Data sources:** dataset name and provider, with web evidence where it can be resolved;⁶ collection method (8-category taxonomy); geographic scope; sample size; access status (public / restricted / proprietary); identification role.
- **Methods:** empirical, structural, descriptive, algorithmic, and measurement approaches used in the paper; method or design family (91-type taxonomy, see Appendix R); validity tests (test name, reported statistic, p -value, result) (Brodeur et al., 2020); how the identifying assumption was defended (statistical test / verbal defense / indirect evidence / not addressed); whether the strategy was deployed, mentioned as an alternative, or proposed for future work.
- **Findings:** reported results, including effect sizes where available, with standard error; null result (affirmatively recorded, not inferred from absence of text); whether the proposed causal mechanism was tested; whether the mediating variable was observed in the data.
- **Limitations:** author-stated constraints, including constraint type from a controlled vocabulary; severity; links to the open questions this limitation blocks.
- **Open questions:** author-stated follow-up questions, including data requirements; question type (mechanism, extension, external validity, policy, methodology, identification).
- **Citation evidence:** references that play a visible role in the citing paper, labelled by function from a taxonomy based on Jurgens et al. (2018) with economics-specific extensions (data or measurement, method justification, compares findings, background, etc.).⁷
- **Replication metadata:** paper-level replication-package status; package or repository URL where resolved; source of the resolution (paper text, OpenAlex, or directed search); and verification evidence for the matched package.

Extracted items are linked to show which data source supports which method, which method produces which finding, and which limitation motivates which open question. Appendix E lists the relationship types. The ten primary citation roles used in the main tables are literature survey, prior evidence, theoretical foundation, method justification, data or measurement, contextual motivation, contribution framing, estimate comparison, identification strategy, and contradiction. Coverage for the 1,608 selected processed NBER working-paper PDFs is reported once, in Section 5: the headline counts are 9,880 methods and design components, 10,498 findings, 7,659 data sources, 6,337 limitations, 4,364 open questions, 1,597 paper-level replication-package status records, and 120,064 role-coded citation links. Citation-role links are reported separately because their natural unit is the local citation context rather than the paper-level information list.

The paper counts differ across layers because the extraction target differs. The method layer records design components used in a paper’s own analysis; the 70 papers without a method record are almost all descriptive review, synthesis, agenda, or commentary-style papers (69 descriptive and one theoretical in the current JSON audit). The finding layer is broader, so many of those papers still contribute extracted claims or summary findings. Conversely, seven papers have no finding record under the current finding prompt; these are mostly method, database, measurement, or overview papers where the useful content is captured in methods, data sources, limitations, or open questions rather than as a discrete result statement.

⁶Dataset resolution uses web search, institutional pages, DOI metadata, OpenAlex, Crossref, and Wikidata where available. Ambiguous cases are marked for review rather than treated as fully resolved.

⁷Cited-work keys are resolved to DOIs via OpenAlex and Crossref where possible. The citing documents are the selected NBER working-paper PDFs; cited works may be working papers, published articles, books, reports, datasets, or other reference types. Unresolved references fall back to author-year identifiers.